Imperial College London

Department of Electronic and Electrical Engineering

Examinations 2026

Digital Signal Processing and Digital Filters

MSc and EEE/EIE PART IV

May, 2026 • Exam Duration: Three Hours

There are FOUR questions on this paper.

Answer Question 1 and any TWO other questions.

Question 1 is worth 40% of the total marks
while each of the other questions is worth 30%.

If necessary, make reasonable assumptions for your solutions and clearly explain them in your answers.

A formula sheet is attached in the next two pages.

Markers: Ayush Bhandari and Tania Stathaki.

Question 1 [40 Marks]

Q1-1 The function $f(t) = \text{sinc}(t) = \frac{\sin(\pi t)}{\pi t}$ defines an ideal bandlimited signal.

What is the Nyquist rate for this signal? **[2 Marks]**

T = 1 seconds.

Q1-2 Let

$$H(z) = \frac{b_0 + b_1 z^{-1} + b_2 z^{-2} + b_3 z^{-3} + b_4 z^{-4}}{1 + a_1 z^{-1} + a_2 z^{-2} + a_3 z^{-3} + a_4 z^{-4}}$$

be a causal LTI filter. Define a second filter $H_R(z) = H(-z)$.

Let $p[n]$ denote the impulse response of $H(z)$ and $q[n]$ denote the impulse response of $H_R(z)$, i.e., $p[n]$ and $q[n]$ are the outputs when the input is $x[n] = \delta[n]$.

(a) Write $H_R(z)$ explicitly in terms of $\{a_k\}$ and $\{b_k\}$. **[3 Marks]**

$$\begin{aligned} H_R(z) = H(-z) &= \frac{b_0 + b_1(-z)^{-1} + b_2(-z)^{-2} + b_3(-z)^{-3} + b_4(-z)^{-4}}{1 + a_1(-z)^{-1} + a_2(-z)^{-2} + a_3(-z)^{-3} + a_4(-z)^{-4}} \\ &= \frac{b_0 - b_1 z^{-1} + b_2 z^{-2} - b_3 z^{-3} + b_4 z^{-4}}{1 - a_1 z^{-1} + a_2 z^{-2} - a_3 z^{-3} + a_4 z^{-4}}. \end{aligned}$$

(b) Show that $q[n] = (-1)^n p[n]$. **[2 Marks]**

If $H(z) = \sum_n p[n] z^{-n}$, then $H_R(z) = H(-z) = \sum_n p[n](-z)^{-n} = \sum_n (-1)^n p[n] z^{-n}$, hence $q[n] = (-1)^n p[n]$.

(c) Deduce that all even-indexed samples of the impulse response are unchanged and all odd-indexed samples change sign, i.e.,

$$q[2m] = p[2m], \qquad q[2m+1] = -p[2m+1].$$

[2 Marks]

From $q[n] = (-1)^n p[n]$, if $n = 2m$ then $(-1)^{2m} = 1$ so $q[2m] = p[2m]$; if $n = 2m+1$ then $(-1)^{2m+1} = -1$ so $q[2m+1] = -p[2m+1]$.

Q1-3 Bilinear Mapping. Consider the analog filter

$$H(s) = \frac{1}{as^2 + bs + c}.$$

The filter is converted into a digital filter using the bilinear transform

$$s = \frac{2}{T} \frac{1 - z^{-1}}{1 + z^{-1}}.$$

Assume that $T = 1$ and that the resulting digital transfer function $H(z)$ has denominator

$$\alpha + \beta z^{-1} + \gamma z^{-2}.$$

i) Express α, β, γ in terms of a, b, c. **[2 Marks]**

Substituting $s = 2\frac{1-z^{-1}}{1+z^{-1}}$ into $H(s)$ and multiplying numerator and denominator by $(1+z^{-1})^2$ gives

$$H(z) = \frac{(1+z^{-1})^2}{4a(1-z^{-1})^2 + 2b(1-z^{-1})(1+z^{-1}) + c(1+z^{-1})^2}.$$

Using

$$(1-z^{-1})^2 = 1-2z^{-1}+z^{-2}, \quad (1+z^{-1})^2 = 1+2z^{-1}+z^{-2}, \quad (1-z^{-1})(1+z^{-1}) = 1-z^{-2},$$

the denominator becomes $(4a+2b+c) + (-8a+2c)z^{-1} + (4a-2b+c)z^{-2}$. Hence

$$\alpha = 4a+2b+c, \qquad \beta = -8a+2c, \qquad \gamma = 4a-2b+c.$$

ii) Suppose that $\gamma = 0$ and that the analog filter has unity DC gain i.e. $H(0) = 1$. Determine c and b in terms of a, and hence express $H(s)$ as a function of a. **[3 Marks]**

From $\gamma = 0$,

$$4a - 2b + c = 0.$$

Unity DC gain implies

$$H(0) = \frac{1}{c} = 1 \quad \Rightarrow \quad c = 1.$$

Substituting into the previous equation,

$$4a - 2b + 1 = 0 \quad \Rightarrow \quad b = 2a + \frac{1}{2}.$$

Therefore the analog filter becomes

$$H(s) = \frac{1}{as^2 + \left(2a+\frac{1}{2}\right)s + 1}.$$

Q1-4 A certain system is defined by its impulse response,

$$x[n] \longrightarrow \boxed{\boxed{h_\lambda[n] = \lambda^{\left\lfloor \frac{n-2}{4} \right\rfloor} u[n-2]}} \longrightarrow y[n], \qquad 0 < \lambda < 1.$$

where $u[n]$ is the unit-step sequence and $\lfloor x \rfloor$ denotes the largest integer less than or equal to x. For example, $\lfloor 2.7 \rfloor = 2$ and $\lfloor 3 \rfloor = 3$.

(a) Compute the z-transform $H_\lambda(z)$ and state the ROC. **[4 Marks]**

Let $m = n - 2 \geq 0$. Then $H_\lambda(z) = \sum_n h_\lambda[n] z^{-n} = z^{-2} \sum_{m=0}^{\infty} \lambda^{\lfloor m/4 \rfloor} z^{-m}$. Write $m = 4r + p$, $p = 0, 1, 2, 3$. Then $\lfloor m/4 \rfloor = r$ and

$$\sum_{m=0}^{\infty} \lambda^{\lfloor m/4 \rfloor} z^{-m} = \sum_{r=0}^{\infty} \sum_{p=0}^{3} \lambda^r z^{-(4r+p)} = (1 + z^{-1} + z^{-2} + z^{-3}) \sum_{r=0}^{\infty} (\lambda z^{-4})^r.$$

Hence

$$H_\lambda(z) = \frac{z^{-2}(1 + z^{-1} + z^{-2} + z^{-3})}{1 - \lambda z^{-4}},$$

with ROC $|\lambda z^{-4}| < 1$, i.e. $|z| > \lambda^{1/4}$.

(b) Show that when $\lambda = 0$,

$$H_0(z) = z^{-2}(1 + z^{-1} + z^{-2} + z^{-3}).$$

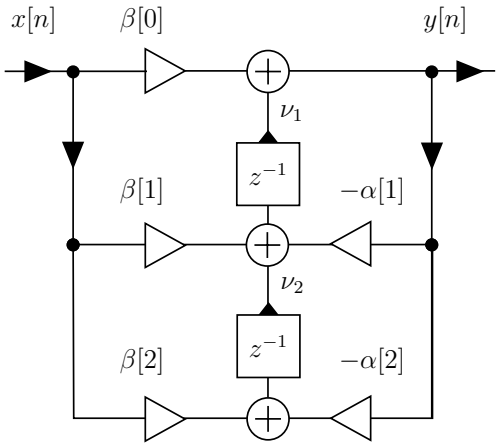
[2 Marks]

Substitute $\lambda = 0$ into the expression of part (a).

(c) Compute the impulse response $h_0[n]$ and its energy $E = \sum_{n=-\infty}^{\infty} |h_0[n]|^2$. **[2 Marks]**

Since $H_0(z) = z^{-2} + z^{-3} + z^{-4} + z^{-5}$, we have $h_0[n] = \delta[n-2] + \delta[n-3] + \delta[n-4] + \delta[n-5]$. Thus $E = \sum_n |h_0[n]|^2 = 1 + 1 + 1 + 1 = 4$.

Q1-5 Consider the 2nd-order biquad shown in the figure, with coefficients $\{\beta[0], \beta[1], \beta[2]\}$ and feedback taps $\{-\alpha[1], -\alpha[2]\}$.

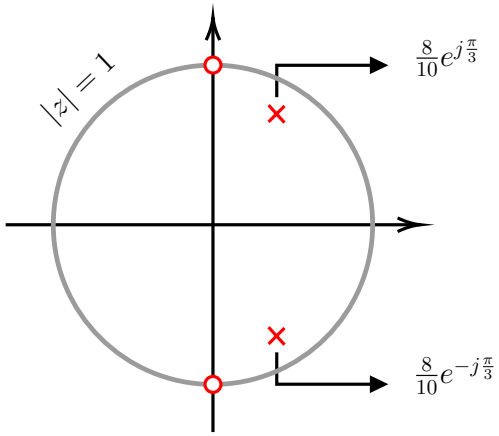


i) Write the transfer function $H(z) = \frac{Y(z)}{X(z)}$ in terms of $\alpha[1], \alpha[2], \beta[0], \beta[1], \beta[2]$. **[2 Marks]**

The structure implements $y[n] + \alpha[1]y[n-1] + \alpha[2]y[n-2] = \beta[0]x[n] + \beta[1]x[n-1] + \beta[2]x[n-2]$, hence

$$H(z) = \frac{\beta[0] + \beta[1]z^{-1} + \beta[2]z^{-2}}{1 + \alpha[1]z^{-1} + \alpha[2]z^{-2}}.$$

ii) Choose specific numerical values of $\alpha[1], \alpha[2], \beta[0], \beta[1], \beta[2]$ such that the filter has the following pole-zero plot:



Write the filter coefficients $\{\alpha[k], \beta[k]\}$. **[2 Marks]**

Zeros at $z = \pm j$ correspond to the numerator polynomial $1 + z^{-2}$, so choose

$$\beta[0] = 1, \qquad \beta[1] = 0, \qquad \beta[2] = 1,$$

giving zeros $z = \pm j$.

Poles at $re^{\pm j\theta}$ with $r = 0.8$, $\theta = \pi/3$ correspond to the denominator polynomial

$$1 - 2r\cos\theta\, z^{-1} + r^2 z^{-2}.$$

Matching $1 + \alpha[1]z^{-1} + \alpha[2]z^{-2}$ gives

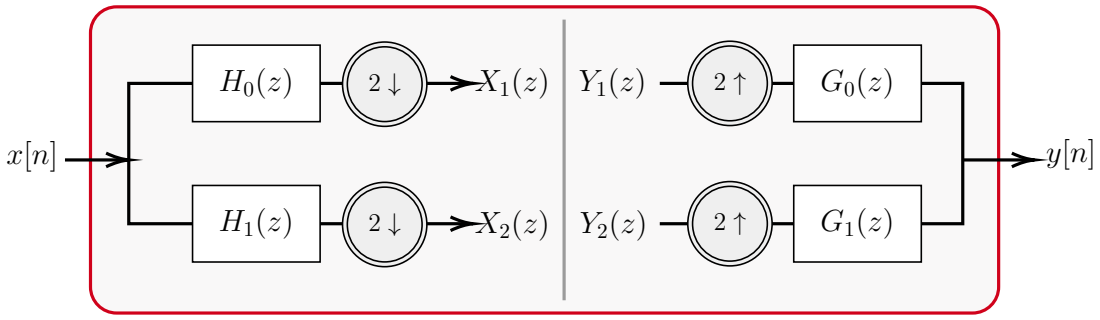
$$\alpha[1] = -2r\cos\theta = -2(0.8)\cos(\pi/3) = -0.8, \qquad \alpha[2] = r^2 = 0.64.$$

Thus the poles are $z = 0.8e^{\pm j\pi/3}$ and the zeros are $z = \pm j$.

iii) Is the filter BIBO stable? Justify. **[2 Marks]**

A causal IIR is BIBO stable iff all poles lie strictly inside the unit circle. Here the poles have magnitude $|z| = 0.8 < 1$, hence the filter is BIBO stable.

Q1-6 Consider the two–channel analysis/synthesis system shown in the figure.



The downsampling and upsampling factors are 2. The analysis low–pass filter is

$$H_0(z) = 1 + 2z^{-1} + z^{-2}.$$

The analysis high–pass filter is $H_1(z)$ and the synthesis filters are $G_0(z)$ and $G_1(z)$.

i) Derive the overall input-output relation and show that it can be written as

$$Y(z) = \frac{1}{2}\Big[T(z)X(z) + A(z)X(-z)\Big].$$

Identify the functions $T(z)$ and $A(z)$. **[4 Marks]**

Using the standard identities for decimation and interpolation by 2, the system output can be written as

$$Y(z) = G_0(z)X_1(z^2) + G_1(z)X_2(z^2),$$

where

$$X_1(z) = \tfrac{1}{2}\Big(H_0(z^{1/2})X(z^{1/2}) + H_0(-z^{1/2})X(-z^{1/2})\Big)\,,$$

$$X_2(z) = \tfrac{1}{2}\Big(H_1(z^{1/2})X(z^{1/2}) + H_1(-z^{1/2})X(-z^{1/2})\Big)\,.$$

Substituting $z \to z^2$ gives

$$X_1(z^2) = \tfrac{1}{2}(H_0(z)X(z) + H_0(-z)X(-z))\,,$$

$$X_2(z^2) = \tfrac{1}{2}(H_1(z)X(z) + H_1(-z)X(-z))\,.$$

Hence

$$Y(z) = \tfrac{1}{2}\Big[(G_0(z)H_0(z)+G_1(z)H_1(z))X(z)+(G_0(z)H_0(-z)+G_1(z)H_1(-z))X(-z)\Big].$$

Therefore

$$T(z) = G_0(z)H_0(z) + G_1(z)H_1(z),$$

$$A(z) = G_0(z)H_0(-z) + G_1(z)H_1(-z).$$

ii) State the conditions for perfect reconstruction. **[2 Marks]**

Perfect reconstruction requires that the aliasing term vanish and that the distortion term be a pure delay and gain. Hence

$$A(z) = 0,$$

and

$$T(z) = c\,z^{-k}$$

for some constant c and integer delay k.

iii) Design $H_1(z)$ such that the analysis filters form a quadrature mirror pair, i.e.

$$H_1(z) = H_0(-z).$$

Compute $H_1(z)$. **[2 Marks]**

Given

$$H_0(z) = 1 + 2z^{-1} + z^{-2},$$

we substitute $z \to -z$:

$$H_1(z) = H_0(-z) = 1 - 2z^{-1} + z^{-2}.$$

iv) Choose synthesis filters $G_0(z)$ and $G_1(z)$ such that aliasing is cancelled. **[2 Marks]**

A standard choice that cancels aliasing is

$$G_0(z) = H_1(-z), \qquad G_1(z) = -H_0(-z).$$

Since

$$H_1(z) = 1 - 2z^{-1} + z^{-2},$$

we obtain

$$G_0(z) = H_1(-z) = 1 + 2z^{-1} + z^{-2},$$
$$G_1(z) = -H_0(-z) = -(1 - 2z^{-1} + z^{-2}).$$

Substituting these expressions into

$$A(z) = G_0(z)H_0(-z) + G_1(z)H_1(-z)$$

shows that $A(z) = 0$.

v) Determine the overall system response $Y(z)$ and verify that reconstruction is achieved up to an LTI distortion proportional to $4z^{-1}(1 + z^{-2})$. **[2 Marks]**

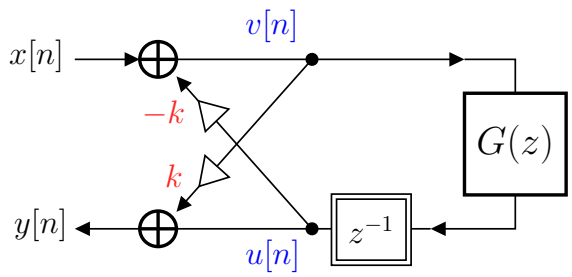
Given $H_0(z) = 1 + 2z^{-1} + z^{-2}$ and $H_1(z) = H_0(-z) = 1 - 2z^{-1} + z^{-2}$. The synthesis filters are $G_0(z) = H_1(-z) = 1 + 2z^{-1} + z^{-2}$ and $G_1(z) = -H_0(-z) = -(1 - 2z^{-1} + z^{-2}) = -1 + 2z^{-1} - z^{-2}$.

The distortion transfer function is $T(z) = G_0(z)H_0(z) + G_1(z)H_1(z)$. Compute $G_0(z)H_0(z) = (1 + 2z^{-1} + z^{-2})^2 = 1 + 4z^{-1} + 6z^{-2} + 4z^{-3} + z^{-4}$ and $G_1(z)H_1(z) = (-1 + 2z^{-1} - z^{-2})(1 - 2z^{-1} + z^{-2}) = -(1 - 2z^{-1} + z^{-2})^2 = -1 + 4z^{-1} - 6z^{-2} + 4z^{-3} - z^{-4}$.

Adding gives $T(z) = 8z^{-1} + 8z^{-3} = 8z^{-1}(1 + z^{-2})$. Since $Y(z) = \frac{1}{2}T(z)X(z)$, the overall response is $Y(z) = 4z^{-1}(1 + z^{-2})X(z)$.

Question 2 [30 Marks]

Q2-1 Consider the discrete-time system given by the filter structure:



The intermediate signals are defined by $v[n]$ and $u[n]$. The output is given by $y[n]$.

Assume $G(z)$ is an N-th order stable allpass filter of the form:

$$G(z) = \frac{z^{-N}A(z^{-1})}{A(z)}, \qquad A(z) = \sum_{n=0}^{N} a[n]\, z^{-n}, \quad a[0] = 1.$$

(a) Provide the time-domain representation of sequences $v[n]$ and $u[n]$. **[2 Marks]**

From the figure: $v[n] = x[n] - k\,u[n]$ and $u[n] = \big(g * v\big)[n-1]$.

(b) Derive $H(z) = \frac{Y(z)}{X(z)}$.
Show that it can be written as

$$H(z) = \frac{k + z^{-1}G(z)}{1 + kz^{-1}G(z)}.$$

[2 Marks]

Since $u[n] = (g * v)[n-1]$, we have $U(z) = z^{-1}G(z)V(z)$. From

$$v[n] = x[n] - ku[n] \quad \Longrightarrow \quad V(z) = X(z) - kU(z) = X(z) - kz^{-1}G(z)V(z),$$

hence

$$V(z)\big(1 + kz^{-1}G(z)\big) = X(z) \quad \Longrightarrow \quad V(z) = \frac{1}{1 + kz^{-1}G(z)}\,X(z).$$

Next,

$$Y(z) = kV(z) + U(z) = \Big(k + z^{-1}G(z)\Big)V(z) = \frac{k + z^{-1}G(z)}{1 + kz^{-1}G(z)}\,X(z),$$

so

$$H(z) = \frac{Y(z)}{X(z)} = \frac{k + z^{-1}G(z)}{1 + kz^{-1}G(z)}.$$

(c) For some polynomial $D(z) = \sum_{n=0}^{N+1} d[n]z^{-n}$, show that $H(z)$ can be expressed in the allpass form:

$$H(z) = z^{-(N+1)}\frac{D(z^{-1})}{D(z)}$$

Write down $D(z)$ explicitly in terms of $A(z)$, $A(z^{-1})$, and k. **[4 Marks]**

Substitute $G(z) = \dfrac{z^{-N}A(z^{-1})}{A(z)}$ into the result of part (b):

$$H(z) = \frac{k + z^{-1}\frac{z^{-N}A(z^{-1})}{A(z)}}{1 + kz^{-1}\frac{z^{-N}A(z^{-1})}{A(z)}} = \frac{kA(z) + z^{-(N+1)}A(z^{-1})}{A(z) + kz^{-(N+1)}A(z^{-1})}.$$

Define

$$D(z) \triangleq A(z) + kz^{-(N+1)}A(z^{-1}).$$

Then

$$D(z^{-1}) = A(z^{-1}) + kz^{N+1}A(z),$$

and the numerator becomes

$$kA(z) + z^{-(N+1)}A(z^{-1}) = z^{-(N+1)}\Big(A(z^{-1}) + kz^{N+1}A(z)\Big) = z^{-(N+1)}D(z^{-1}).$$

Therefore,

$$H(z) = \frac{z^{-(N+1)}D(z^{-1})}{D(z)}.$$

(d) By comparing coefficients, derive $d[n]$ in terms of $a[n]$ and k, for $n = 0, 1, \ldots, N+1$. **[4 Marks]**

Write $A(z) = \sum_{n=0}^{N} a[n]z^{-n}$. Then

$$z^{-(N+1)}A(z^{-1}) = z^{-(N+1)}\sum_{n=0}^{N} a[n]z^{n} = \sum_{n=0}^{N} a[n]\, z^{-(N+1-n)}.$$

Hence

$$D(z) = A(z) + kz^{-(N+1)}A(z^{-1}) = \sum_{n=0}^{N} a[n]z^{-n} + k\sum_{n=0}^{N} a[n]z^{-(N+1-n)}.$$

Matching $D(z) = \sum_{n=0}^{N+1} d[n]z^{-n}$, we obtain

$$d[0] = a[0] = 1, \qquad d[N+1] = ka[0] = k,$$

and for $1 \le n \le N$,

$$\boxed{d[n] = a[n] + k\, a[N+1-n].}$$

(e) Suppose you are instead given $\{d[n]\}_{n=0}^{N+1}$. Derive how to recover k and then $\{a[n]\}_{n=0}^{N}$ from $\{d[n]\}$, assuming $|k| \neq 1$. **[4 Marks]**

From part (c), $d[N+1] = k$, so

$$\boxed{k = d[N+1].}$$

For $1 \le n \le N$, use the paired equations

$$d[n] = a[n] + ka[N+1-n], \qquad d[N+1-n] = a[N+1-n] + ka[n].$$

Form $d[n] - kd[N+1-n]$:

$$d[n] - kd[N+1-n] = (a[n] + ka[N+1-n]) - k(a[N+1-n] + ka[n]) = (1-k^2)a[n].$$

Assuming $|k| \neq 1$,

$$\boxed{a[n] = \frac{d[n] - k\, d[N+1-n]}{1-k^2}}, \qquad 1 \le n \le N,$$

with $a[0] = 1$ given.

(f) Assuming $G(z)$ is stable, state a necessary and sufficient condition on k for $H(z)$ to be stable. **[2 Marks]**

The transfer function is

$$H(z) = \frac{k + z^{-1}G(z)}{1 + kz^{-1}G(z)}.$$

With $G(z)$ stable and allpass, stability of the closed-loop section is governed by the reflection coefficient k. The necessary and sufficient condition is

$$\boxed{|k| < 1.}$$

Q2-2 The type-II Discrete Cosine Transform (DCT) of a length-N sequence $x[n]$ is defined as

$$X_C[k] = \sum_{n=0}^{N-1} x[n] \cos\left(\frac{2\pi(2n+1)k}{4N}\right), \qquad k = 0, 1, \ldots, N-1.$$

Let

$$X(z) = \sum_{n=0}^{N-1} x[n] z^{-n}$$

denote the z-transform of the sequence $x[n]$. Show that the DCT coefficients can be written as

$$X_C[k] = \Re\left\{ e^{-j\frac{\pi k}{2N}} X\left(e^{j\frac{\pi k}{N}}\right)\right\}.$$

[6 Marks]

Starting from the definition

$$X_C[k] = \sum_{n=0}^{N-1} x[n] \cos\left(\frac{2\pi(2n+1)k}{4N}\right) = \sum_{n=0}^{N-1} x[n] \cos\left(\frac{\pi(2n+1)k}{2N}\right).$$

Using

$$\cos\theta = \Re\{e^{j\theta}\},$$

we obtain

$$X_C[k] = \Re\left\{ \sum_{n=0}^{N-1} x[n] e^{j\frac{\pi(2n+1)k}{2N}} \right\}.$$

Rewrite the exponential term:

$$e^{j\frac{\pi(2n+1)k}{2N}} = e^{j\frac{\pi k}{2N}} e^{j\frac{\pi nk}{N}}.$$

Hence

$$X_C[k] = \Re\left\{ e^{j\frac{\pi k}{2N}} \sum_{n=0}^{N-1} x[n] e^{j\frac{\pi nk}{N}} \right\}.$$

Now note that the z-transform evaluated on the unit circle is

$$X(e^{j\omega}) = \sum_{n=0}^{N-1} x[n] e^{-j\omega n}.$$

Therefore

$$\sum_{n=0}^{N-1} x[n]e^{j\frac{\pi nk}{N}} = X\left(e^{-j\frac{\pi k}{N}}\right).$$

Substituting gives

$$X_C[k] = \Re\left\{e^{j\frac{\pi k}{2N}} X\left(e^{-j\frac{\pi k}{N}}\right)\right\}.$$

Since the real part is invariant under complex conjugation,

$$\Re\{z\} = \Re\{z^*\},$$

we obtain

$$X_C[k] = \Re\left\{e^{-j\frac{\pi k}{2N}} X\left(e^{j\frac{\pi k}{N}}\right)\right\}.$$

Hence the DCT coefficients can be interpreted as the real part of the z-transform evaluated on the unit circle with a half-sample phase shift:

$$X_C[k] = \Re\left\{e^{-j\frac{\pi k}{2N}} X\left(e^{j\frac{\pi k}{N}}\right)\right\}.$$

Q2-3 Let $h[n]$ be the impulse response of an FIR filter with frequency response

$$H(e^{j\omega}) = \sum_n h[n]e^{-j\omega n}.$$

The group delay is defined as $\tau_g(\omega) = -\dfrac{d}{d\omega}\arg H(e^{j\omega})$.

(a) Show that

$$\frac{d}{d\omega}H(e^{j\omega}) = -j\sum_n nh[n]e^{-j\omega n}.$$

[3 Marks]

Differentiate the DTFT: $\frac{d}{d\omega}H(e^{j\omega}) = \sum_n h[n]\frac{d}{d\omega}(e^{-j\omega n})$. Since $\frac{d}{d\omega}e^{-j\omega n} = -jne^{-j\omega n}$, we obtain $\frac{d}{d\omega}H(e^{j\omega}) = -j\sum_n n\,h[n]e^{-j\omega n}$.

(b) Using the identity

$$\tau_g(\omega) = \frac{\Re\{H^*(e^{j\omega})\, j\,\frac{d}{d\omega}H(e^{j\omega})\}}{|H(e^{j\omega})|^2},$$

show that for a linear-phase FIR filter

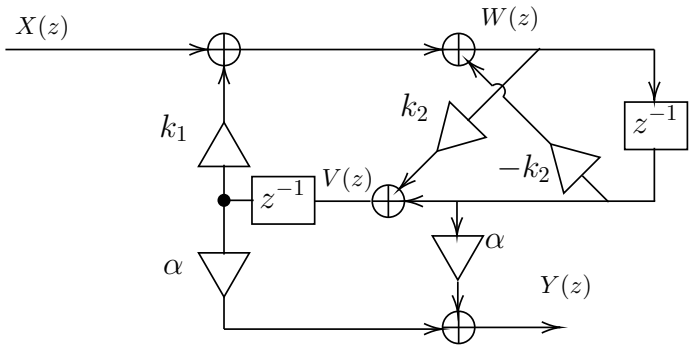
$$\tau_g = \frac{\sum_n n|h[n]|^2}{\sum_n |h[n]|^2}.$$

[3 Marks]

From part (a), $j\frac{d}{d\omega}H(e^{j\omega}) = \sum_n n\,h[n]e^{-j\omega n}$. Hence $H^*(e^{j\omega})\,j\frac{d}{d\omega}H(e^{j\omega}) = (\sum_m h[m]e^{j\omega m})(\sum_n n\,h[n]e^{-j\omega n})$. Taking the real part and using the symmetry of a linear-phase FIR filter, cross-terms cancel and only $m = n$ terms remain, giving $\Re\{\cdot\} = \sum_n n|h[n]|^2$. Also $|H(e^{j\omega})|^2 = \sum_n |h[n]|^2$. Substitution yields $\tau_g = \frac{\sum_n n|h[n]|^2}{\sum_n |h[n]|^2}$, i.e., the energy centroid of $h[n]$.

Question 3 [30 Marks]

Q3-1 Let $H(z) = \frac{Y(z)}{X(z)}$ be a system represented in the diagram below



Suppose that $k_1 k_2 = 1$. Find the zeros of the filter $H(z)$. **[8 Marks]**

Using signal-flow (node) equations for the diagram and eliminating internal variables, the transfer function for independent k_1, k_2 is

$$H(z) = \frac{Y(z)}{X(z)} = \frac{\alpha z^{-1}\big(1 + k_2 + z^{-1}\big)}{1 + \big(k_2 - k_1 k_2\big)z^{-1} - k_1 z^{-2}}.$$

(We will now use the given constraint $k_1 k_2 = 1$.)

Since $k_1 k_2 = 1$, we have $k_1 = \frac{1}{k_2}$, and therefore

$$1 + \big(k_2 - k_1 k_2\big)z^{-1} - k_1 z^{-2} = 1 + \big(k_2 - 1\big)z^{-1} - \frac{1}{k_2} z^{-2}.$$

Hence the transfer function under the constraint $k_1 k_2 = 1$ is

$$\boxed{H(z) = \frac{\alpha z^{-1}\big(1 + k_2 + z^{-1}\big)}{1 + \big(k_2 - 1\big)z^{-1} - \frac{1}{k_2} z^{-2}}}.$$

For pole/zero inspection it is convenient to multiply numerator and denominator by z^2:

$$H(z) = \alpha \frac{(1 + k_2)z + 1}{z^2 + (k_2 - 1)z - \frac{1}{k_2}}.$$

Thus the zeros are $z = 0$ (from the z^{-1} factor) and $z = -\frac{1}{1+k_2}$ (for $k_2 \neq -1$).

Q3-2 A filter, with input $x[n]$ and output $y[n]$, is defined by the difference equation

$$y[n] + a_1 y[n-1] + a_2 y[n-2] = a_2 x[n] + a_1 x[n-1] + x[n-2],$$

where $a_1, a_2 \in \mathbb{R}$. Assume the system is stable.

(a) Determine the system function $H(z)$ of the filter. **[3 Marks]**

Taking the z-transform (zero initial conditions),

$$Y(z)(1 + a_1 z^{-1} + a_2 z^{-2}) = X(z)(a_2 + a_1 z^{-1} + z^{-2}).$$

Hence

$$H(z) = \frac{Y(z)}{X(z)} = \frac{a_2 + a_1 z^{-1} + z^{-2}}{1 + a_1 z^{-1} + a_2 z^{-2}}.$$

(b) Show that the filter is an all-pass filter, i.e. $|H(e^{j\omega})| = 1$ for all ω. **[2 Marks]**

Let $A(z) = 1 + a_1 z^{-1} + a_2 z^{-2}$. Then

$$H(z) = \frac{z^{-2} A(z^{-1})}{A(z)}.$$

For $z = e^{j\omega}$,

$$H(e^{j\omega}) = \frac{e^{-j2\omega} A(e^{-j\omega})}{A(e^{j\omega})}.$$

Since $A(e^{-j\omega}) = A^*(e^{j\omega})$ (real coefficients),

$$|H(e^{j\omega})| = \frac{|A(e^{j\omega})|}{|A(e^{j\omega})|} = 1.$$

(c) Determine the locations of the poles and zeros of $H(z)$ in terms of a_1 and a_2, and briefly describe the relationship between them. **[3 Marks]**

The poles are the roots of

$$1 + a_1 z^{-1} + a_2 z^{-2} = 0$$

or equivalently

$$z^2 + a_1 z + a_2 = 0.$$

The zeros are the roots of

$$a_2 + a_1 z^{-1} + z^{-2} = 0$$

or

$$a_2 z^2 + a_1 z + 1 = 0.$$

Thus if z_p is a pole then $z_z = 1/z_p$ is a zero. Hence poles and zeros occur in reciprocal pairs.

(d) Let $\tau_g(\omega)$ denote the group delay of the filter.
Determine conditions on a_1 and a_2 such that

$$\tau_g(0) = \tau_g\left(\frac{\pi}{4}\right).$$

[4 Marks]

If $a_1 = 0$,

$$H(z) = \frac{a_2 + z^{-2}}{1 + a_2 z^{-2}},$$

which depends only on z^{-2}. On the unit circle $z = e^{j\omega}$,

$$H(e^{j\omega}) = G(e^{j2\omega})$$

for some function $G(\cdot)$. Hence the response is periodic with period $\pi/2$:

$$H(e^{j(\omega+\pi/2)}) = H(e^{j\omega}).$$

This symmetry implies that the phase slope (group delay) at $\omega = 0$ equals that at $\omega = \pi/4$. Thus the condition is

$$a_1 = 0.$$

(e) Under the condition obtained in part (d), sketch qualitatively the pole–zero pattern of the filter and explain how the symmetry of this pattern leads to the observed behaviour of the group delay. **[2 Marks]**

If $a_1 = 0$, the poles satisfy $z^2 + a_2 = 0$, so $z = \pm j\sqrt{a_2}$, and the zeros are their reciprocals $z = \pm j/\sqrt{a_2}$. Thus poles and zeros lie symmetrically on the imaginary axis in reciprocal pairs, producing a frequency response symmetric about $\omega = \pi/4$, which leads to the equality of group delay at $\omega = 0$ and $\omega = \pi/4$.

Q3-3 Let the system $G(z)$ be defined as

$$G(z) = \frac{5 - z^{-1}}{3 - (1+\alpha)z^{-1} + z^{-2}}, \qquad \alpha \in \mathbb{R}.$$

Find all α such that $G(z)$ has two *distinct real* poles inside the unit circle. **[8 Marks]**

The poles are the roots of the denominator:

$$3 - (1+\alpha)z^{-1} + z^{-2} = 0 \iff 3z^2 - (1+\alpha)z + 1 = 0.$$

For two distinct real poles we require the discriminant to be positive:

$$\Delta = (1+\alpha)^2 - 12 > 0 \iff |1+\alpha| > 2\sqrt{3}.$$

To have both poles inside the unit circle, apply Jury conditions to $3z^2 - (1+\alpha)z + 1$ (with $a_0 = 3, a_1 = -(1+\alpha), a_2 = 1$):

$$|a_2| < a_0 \iff 1 < 3 \quad \text{(always true)},$$

$$a_0 + a_1 + a_2 > 0 \iff 3 - (1+\alpha) + 1 > 0 \iff \alpha < 3,$$

$$a_0 - a_1 + a_2 > 0 \iff 3 + (1+\alpha) + 1 > 0 \iff \alpha > -5.$$

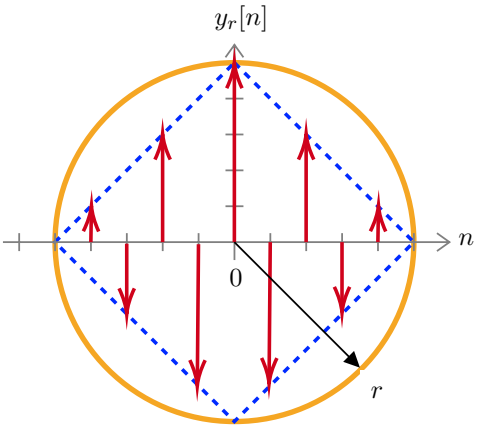
Hence stability requires $-5 < \alpha < 3$. Combining with $|1+\alpha| > 2\sqrt{3}$ yields

$$\alpha \in (-5,\ -1-2\sqrt{3}) \ \cup \ (-1+2\sqrt{3},\ 3).$$

Therefore the set of α is $\boxed{(-5,\ -1-2\sqrt{3}) \cup (-1+2\sqrt{3},\ 3)}$.

Question 4 [30 Marks]

Q4-1 Let $r > 0$ be a real-valued number.



Calculate the energy of the signal $y_r[n]$. **[6 Marks]**

Show that when r is an integer, the energy reduces to:

$$E(r) = \frac{2r^3 + r}{3}.$$

[2 Marks]

Hint: Use the formulas to simplify your calculations:
$\sum_{k=0}^{K-1} k = \frac{K(K-1)}{2}$ and $\sum_{k=0}^{K-1} k^2 = \frac{K(K-1)(2K-1)}{6}$.

From the diamond constraint in the diagram (an ℓ_1 ball of radius r), the vertical extent at index n satisfies

$$|n| + |y_r[n]| \le r.$$

Hence the (nonnegative) amplitude of the red arrow is

$$y_r[n] = \begin{cases} r - |n|, & |n| \le \lfloor r \rfloor, \\ 0, & \text{otherwise.} \end{cases}$$

Energy. The energy of the discrete-time signal is

$$E(r) \triangleq \sum_{n=-\infty}^{\infty} \big(y_r[n]\big)^2 = \sum_{n=-\lfloor r \rfloor}^{\lfloor r \rfloor} (r - |n|)^2.$$

By even symmetry,

$$E(r) = r^2 + 2\sum_{n=1}^{\lfloor r \rfloor} (r - n)^2.$$

Closed form in terms of r. Let $m \triangleq \lfloor r \rfloor$ (so $r \in [m, m+1)$). Then

$$E(r) = r^2 + 2\sum_{n=1}^{m} (r - n)^2 = r^2 + 2\sum_{k=0}^{m-1} (k + (r - m))^2,$$

where we set $k = m - n$. Using

$$\sum_{k=0}^{m-1} k = \frac{m(m-1)}{2}, \qquad \sum_{k=0}^{m-1} k^2 = \frac{(m-1)m(2m-1)}{6},$$

we obtain

$$\sum_{k=0}^{m-1} (k + (r-m))^2 = \frac{(m-1)m(2m-1)}{6} + (r-m)\, m(m-1) + m\,(r-m)^2.$$

Therefore, the energy is

$$\boxed{E(r) \;=\; r^2 + \frac{(m-1)m(2m-1)}{3} + 2(r-m)\, m(m-1) + 2m\,(r-m)^2, \quad m = \lfloor r \rfloor.}$$

Special case: integer radius. If $r \in \mathbb{Z}$, then $m = r$ and $r - m = 0$, so

$$\boxed{E(r) = \frac{2r^3 + r}{3}.}$$

Q4-2 Let $S(z)$ be the z-transform of a sequence $s[k]$, and suppose that

$$(1 - z^{-1})S(z) = 1 + z^{-1}.$$

i) Determine the sequence $s[k]$. **[3 Marks]**

Rearranging,

$$S(z) = \frac{1 + z^{-1}}{1 - z^{-1}}.$$

Write

$$S(z) = \frac{z^{-1}}{1 - z^{-1}} - \frac{z}{1 - z}.$$

Recognizing standard z-transform pairs,

$$\frac{z^{-1}}{1 - z^{-1}} \leftrightarrow u[k-1], \qquad \frac{z}{1 - z} \leftrightarrow u[-1-k].$$

Hence

$$s[k] = u[k-1] - u[-1-k].$$

Equivalently,

$$s[k] = \begin{cases} 1, & k > 0, \\ 0, & k = 0, \\ -1, & k < 0. \end{cases}$$

ii) Evaluate the frequency response $S(e^{j\omega})$. **[2 Marks]**

Substitute $z = e^{j\omega}$,

$$S(e^{j\omega}) = \frac{1 + e^{-j\omega}}{1 - e^{-j\omega}}.$$

Multiply numerator and denominator by $e^{j\omega/2}$,

$$S(e^{j\omega}) = \frac{e^{j\omega/2} + e^{-j\omega/2}}{e^{j\omega/2} - e^{-j\omega/2}} = \frac{2\cos(\omega/2)}{2j\sin(\omega/2)} = -j\cot\left(\frac{\omega}{2}\right).$$

iii) Let $\overline{h}[n] = h[-n]$ and define

$$h_E[n] = \frac{1}{2}\big(h[n] + \overline{h}[n]\big), \qquad h_O[n] = \frac{1}{2}\big(h[n] - \overline{h}[n]\big).$$

Show that

$$\Re(H(e^{j\omega})) = H_E(e^{j\omega}), \qquad \Im(H(e^{j\omega})) = -jH_O(e^{j\omega}).$$

[2 Marks]

The sequence $h[n]$ can be written as

$$h[n] = h_E[n] + h_O[n],$$

where $h_E[n]$ is even and $h_O[n]$ is odd. The Fourier transform of an even sequence is real, and the Fourier transform of an odd sequence is purely imaginary. Hence

$$\Re(H(e^{j\omega})) = H_E(e^{j\omega}), \qquad \Im(H(e^{j\omega})) = -jH_O(e^{j\omega}).$$

iv) Suppose that $h[n]$ is causal. Show that

$$h_O[n] = h_E[n]\, s[n].$$

Hence show that

$$\Im(H(e^{j\omega})) = -\Re(H(e^{j\omega})) \odot \cot\left(\frac{\omega}{2}\right),$$

where $\odot$ denotes circular convolution. **[3 Marks]**

For a causal sequence $h[n]$, we have $h[n] = 0$ for $n < 0$. Using the definitions of even and odd parts,

$$h_O[n] = \frac{h[n] - h[-n]}{2}, \qquad h_E[n] = \frac{h[n] + h[-n]}{2}.$$

Since $h[-n] = 0$ for $n > 0$, it follows that

$$h_O[n] = h_E[n]\, s[n].$$

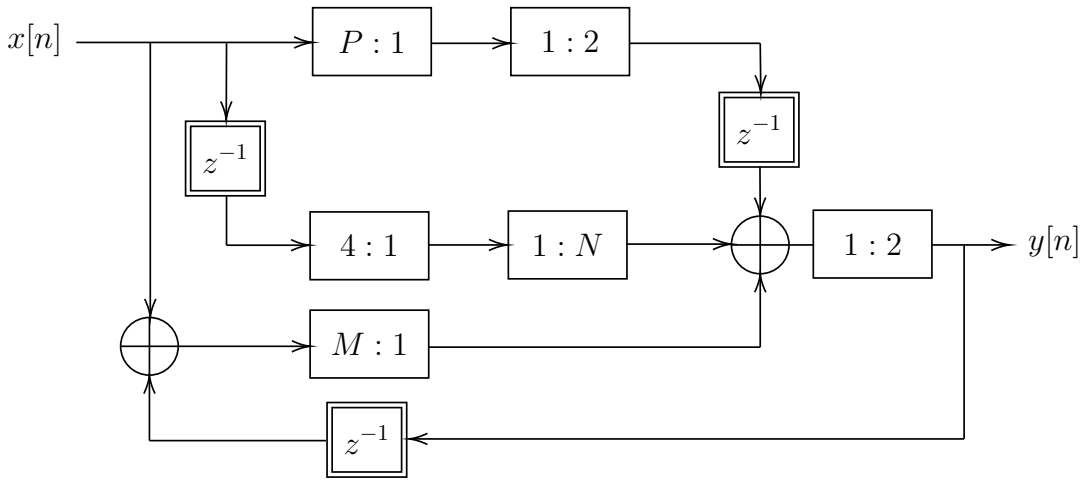
Taking the Fourier transform,

$$H_O(e^{j\omega}) = H_E(e^{j\omega}) \odot S(e^{j\omega}).$$

Using $S(e^{j\omega}) = -j\cot(\omega/2)$ and $H_E(e^{j\omega}) = \Re(H(e^{j\omega}))$,

$$\Im(H(e^{j\omega})) = -jH_O(e^{j\omega}) = -\Re(H(e^{j\omega})) \odot \cot\left(\frac{\omega}{2}\right).$$

Q4-3 Let $x\,[n]$ be the input and $y\,[n]$ the output of a multirate system as defined in the diagram below



i) Calculate M, N, P such that all computations above are valid. **[2 Marks]**

i) Find M, N, P so that all computations are valid.

Interpret each block $L:R$ as a rate change by the factor $\frac{R}{L}$ (down by L, up by R). The three branches feeding the right adder must operate at the same rate.

Top branch: $x[n] \xrightarrow{P:1} \xrightarrow{1:2}$ has rate factor $\frac{1}{P} \cdot 2 = \frac{2}{P}$.

Middle branch: $x[n] \xrightarrow{z^{-1}} \xrightarrow{4:1} \xrightarrow{1:N}$ has rate factor $\frac{1}{4} \cdot N = \frac{N}{4}$.

Bottom branch: (left-adder output) $\xrightarrow{M:1}$ has rate factor $\frac{1}{M}$.

Equating rates: $\frac{2}{P} = \frac{N}{4} = \frac{1}{M}$. The smallest integer solution is

$$\boxed{P = 4, \quad N = 2, \quad M = 2.}$$

ii) Derive the difference equation of the system. **[4 Marks]**

ii) Difference equation / input–output relation.

With $P = 4, N = 2, M = 2$, define the signal at the output of the right adder (i.e., input to the final $1:2$ block) as $r[k]$; this sequence runs at half the input rate. The final block $1:2$ upsamples by 2, hence

$$y[2k] = r[k], \qquad y[2k+1] = 0.$$

Left adder output is $v[n] = x[n] + y[n-1]$. The bottom branch applies $M:1$ with $M = 2$ (downsample by 2), hence it contributes $v[2k] = x[2k] + y[2k-1] = x[2k]$ since $y[2k-1] = 0$.

Top branch: $P:1$ with $P = 4$ (downsample by 4) gives $x[4m]$, then $1:2$ upsamples by 2 to the k-grid, giving a sequence $t[k]$ with

$$t[2m] = x[4m], \qquad t[2m+1] = 0.$$

This branch then passes through a unit delay (at the k-rate), contributing $t[k-1]$.

Middle branch: the delay gives $x[n-1]$, then $4:1$ (down by 4) gives $x[4m-1]$, then $1:2$ upsamples to the k-grid, giving $m[k]$ with

$$m[2m] = x[4m-1], \qquad m[2m+1] = 0.$$

Therefore, at the right adder (at the k-rate),

$$r[k] = t[k-1] + m[k] + x[2k].$$

Convert to $y[n]$ using $y[2k] = r[k]$, $y[2k+1] = 0$:

$$y[4m] = r[2m] = t[2m-1] + m[2m] + x[4m] = 0 + x[4m-1] + x[4m],$$

$$y[4m+2] = r[2m+1] = t[2m] + m[2m+1] + x[4m+2] = x[4m] + 0 + x[4m+2],$$

and $y[2k+1] = 0$. Hence the output is

$$\boxed{y[n] = \begin{cases} x[n] + x[n-1], & n \equiv 0 \pmod 4, \\ x[n] + x[n-2], & n \equiv 2 \pmod 4, \\ 0, & n \text{ odd}. \end{cases}}$$

iii) Design a generalised linear phase filter $H(z)$ that cancels out a harmonic of frequency $\omega_0 = \frac{\pi}{2}$. Additionally, ensure that the gain at $\omega_1 = \frac{\pi}{4}$ is 1 and the gain at $\omega_2 = \frac{3\pi}{4}$ is 5.

[6 Marks]

iii) Design a generalized linear-phase filter $H(z)$.

Use a Type-I linear-phase FIR of length 5 with symmetric coefficients

$$h[0] = h[4] = a, \quad h[1] = h[3] = b, \quad h[2] = c.$$

Then

$$H(e^{j\omega}) = e^{-j2\omega}\Big(c + 2b\cos\omega + 2a\cos 2\omega\Big),$$

so the (real) amplitude term is $A(\omega) = c + 2b\cos\omega + 2a\cos 2\omega$.

Cancel $\omega_0 = \pi/2$: $A(\pi/2) = c + 0 + 2a\cos\pi = c - 2a = 0 \Rightarrow c = 2a$.

Impose $|H(e^{j\pi/4})| = 1$ and $|H(e^{j3\pi/4})| = 5$. Since $e^{-j2\omega}$ is a pure phase, this is equivalent to

$$A(\pi/4) = 1, \qquad A(3\pi/4) = 5.$$

Using $\cos(\pi/4) = \frac{\sqrt{2}}{2}$ and $\cos(2\cdot\pi/4) = \cos(\pi/2) = 0$ (same for $3\pi/4$),

$$A(\pi/4) = 2a + \sqrt{2}\,b = 1, \qquad A(3\pi/4) = 2a - \sqrt{2}\,b = 5.$$

Solving gives $a = \frac{3}{2}$, $b = -\sqrt{2}$, $c = 3$. Therefore

$$\boxed{H(z) = \frac{3}{2}(1 + z^{-4}) - \sqrt{2}\,(z^{-1} + z^{-3}) + 3z^{-2}.}$$

This $H(z)$ is linear phase (symmetric $h[n]$), satisfies $H(e^{j\pi/2}) = 0$, and has gains $|H(e^{j\pi/4})| = 1$, $|H(e^{j3\pi/4})| = 5$.